FUNDAMENTALS OF DATABASES

CLINIC MANAGEMENT SYSTEM

Database Project Report

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PROJECT OVERVIEW

The Clinic Management System is designed to manage the main operations of a medical clinic.
The system stores and organizes:

- Departments
- Clinics
- Doctors
- Patients
- Appointments
- Diagnoses
- Appointment costs and payments

The main purpose of the system is to support appointment scheduling and store patient medical history in a structured database.

PROJECT REQUIREMENTS

The project required designing and implementing a relational database system that includes:

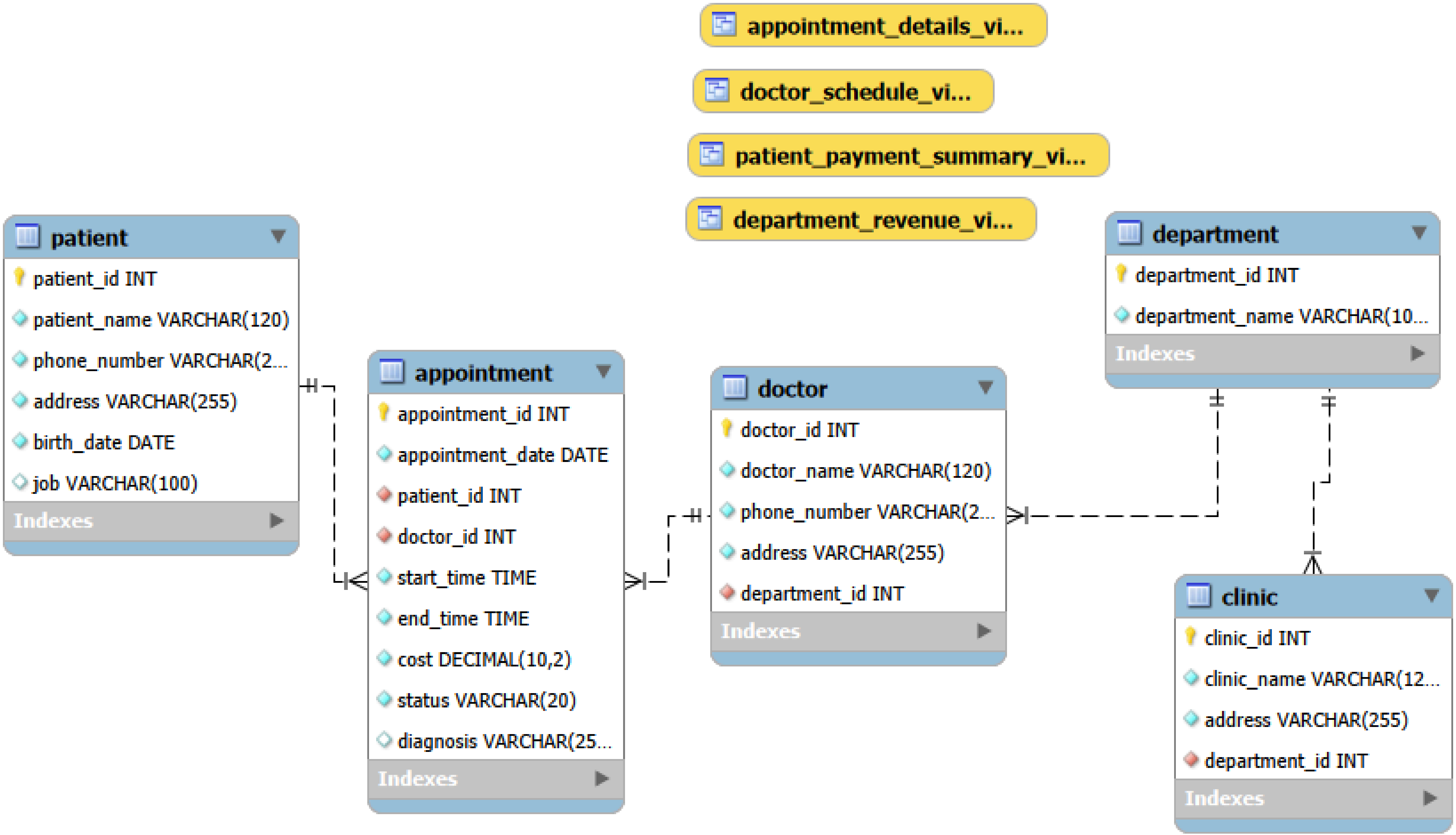
- Conceptual design of the main entities
- ER diagram with cardinality and participation constraints
- Relational schema with tables, attributes, primary keys, and foreign keys
- SQL DDL statements to create tables and views
- SQL DML statements to insert test data
- At least 10 records per table
- Simple and complex SQL queries
- Bonus application/UI that reads from and writes to the database

MAIN ENTITIES

Entity	Description
Department	Stores medical departments such as Cardiology and Gastroenterology
Clinic	Stores clinic branches and their addresses
Doctor	Stores doctor information and department assignment
Patient	Stores patient personal and contact information
Appointment	Stores visit date, time, doctor, patient, status, diagnosis, and cost

ENTITIES AND ATTRIBUTES

Entity	Attributes
Department	department_id, department_name
Clinic	clinic_id, clinic_name, address, department_id
Doctor	doctor_id, doctor_name, phone_number, address, department_id
Patient	patient_id, patient_name, phone_number, address, birth_date, job
Appointment	appointment_id, appointment_date, patient_id, doctor_id, start_time, end_time, cost, status, diagnosis


appointment_details_vi...
doctor_schedule_vi...
patient_payment_summary_vi...
department_revenue_vi...
patient
patient_id INT
patient_name VARCHAR(120)
phone_number VARCHAR(2...
address VARCHAR(255)
birth_date DATE
job VARCHAR(100)
Indexes
appointment
appointment_id INT
appointment_date DATE
patient_id INT
doctor_id INT
start_time TIME
end_time TIME
cost DECIMAL(10,2)
status VARCHAR(20)
diagnosis VARCHAR(25...
Indexes
doctor
doctor_id INT
doctor_name VARCHAR(120)
phone_number VARCHAR(2...
address VARCHAR(255)
department_id INT
Indexes
department
department_id INT
department_name VARCHAR(10...
Indexes
clinic
clinic_id INT
clinic_name VARCHAR(12...
address VARCHAR(255)
department_id INT
Indexes

CARDINALITY AND PARTICIPATION CONSTRAINTS

Relationship	Cardinality	Participation
Department to Clinic	One-to-Many	Clinic has total participation; Department has partial participation
Department to Doctor	One-to-Many	Doctor has total participation; Department has partial participation
Patient to Appointment	One-to-Many	Appointment has total participation; Patient has partial participation
Doctor to Appointment	One-to-Many	Appointment has total participation; Doctor has partial participation

RELATIONAL SCHEMA

Relational schema

Department	(department_id PK, department_name UNIQUE
Clinic	clinic_id PK, clinic_name, address, department_id FK
Doctor	doctor_id PK, doctor_name, phone_number UNIQUE, address, department_id FK
Patient	patient_id PK, patient_name, phone_number UNIQUE, address, birth_date, job
Appointment	appointment_id PK, appointment_date, patient_id FK, doctor_id FK, start_time,

SQL CONSTRAINTS

The database uses several constraints to maintain data accuracy and consistency:

- Primary keys: uniquely identify each record.
- Foreign keys: connect related tables.
- NOT NULL: constraints ensure required fields are entered.
- * UNIQUE:constraints prevent duplicate phone numbers and duplicate clinic entries.
- CHECK:constraint ensures appointment cost is not negative.
- CHECK:constraint ensures appointment end time is after start time.
- CHECK constraint limits appointment status to valid values.

SQL VIEWS

VIEW	PURPOSE
appointment_details_view	| Shows appointment details with patient, doctor, and department names |
patient_payment_summary_view	Shows appointment count and total payment for each patient
Doctor_schedule_view	Shows each doctor's appointment schedule
department_revenue_view	Shows total revenue generated by each department

QUERY 1: FATTY LIVER DIAGNOSIS

Requirement
List the names of patients who were diagnosed with fatty liver in the last year.

SQL Purpose:
The query searches appointment diagnoses containing “fatty liver” and filters appointments from the last year.

```
-- 1. Patients diagnosed with fatty liver within the past year
SELECT DISTINCT p.patient_name
FROM Patient p
JOIN Appointment a ON p.patient_id = a.patient_id
WHERE (LOWER(a.diagnosis) LIKE '%fatty liver%')
  AND a.appointment_date >= DATE_SUB(CURDATE(), INTERVAL 1 YEAR);
```

Result:
Mostafa Mahmoud
Farah Ghaly

```
Mostafa Mahmoud
Farah Ghaly
```

QUERY 2: CARDIOLOGY CLINIC ADDRESSES

Requirement:
List the addresses of cardiology clinics

SQL Purpose:

The query joins the Clinic and Department tables and filters records where the department name is Cardiology.

```
-- 2. Addresses of all cardiology clinics
SELECT c.clinic_name, c.address
FROM Clinic c
JOIN Department d ON c.department_id = d.department_id
WHERE d.department_name = 'Cardiology';
```

Result:

Heart Care Clinic	12 Nile Street, Dokki, Giza
Cardiology Follow-up Clinic	45 Tahrir Street, Downtown Cairo

QUERY 3: TOTAL MONEY PAID BY PATIENT ID 12527

Requirement:
List the total money paid by patient ID 12527 in the last three years.

SQL Purpose:
The query joins Patient and Appointment tables, filters by patient ID, excludes cancelled appointments, and calculates the total cost.

```
-- 3. Total money paid by patient 12527 in the last three years
SELECT p.patient_id, p.patient_name, SUM(a.cost) AS
total_paid_last_three_years
FROM Patient p
JOIN Appointment a ON p.patient_id = a.patient_id
WHERE p.patient_id = 12527
  AND a.status <> 'cancelled'
  AND a.appointment_date >= DATE_SUB(CURDATE(), INTERVAL 3 YEAR);
```

Result:

patient_id	patient_name	total_paid_last_three_years
12527	Mostafa Mahmoud	2100.00

TEST DATA

```sql
(7, 'Ophthalmology'),
(8, 'ENT'),
(9, 'Internal Medicine'),
(10, 'Dental Medicine');

INSERT INTO Clinic (clinic_id, clinic_name, address, department_id) VALUES
(101, 'Heart Care Clinic', '12 Nile Street, Dokki, Giza', 1),
(102, 'Cardiology Follow-up Clinic', '45 Tahrir Street, Downtown Cairo', 1),
(103, 'Digestive Health Clinic', '18 Abbas El Akkad Street, Nasr City, Cairo', 2),
(104, 'Liver and Nutrition Clinic', '22 El Merghany Street, Heliopolis, Cairo', 2),
(105, 'Skin Care Clinic', '9 Makram Ebeid Street, Nasr City, Cairo', 3),
(106, 'Child Wellness Clinic', '31 Syria Street, Mohandessin, Giza', 4),
(107, 'Bone and Joint Clinic', '50 Gameat El Dewal Street, Mohandessin, Giza', 5),
(108, 'Brain and Nerve Clinic', '77 El Merghany Street, Heliopolis, Cairo', 6),
(109, 'Vision Clinic', '14 El Batal Ahmed Street, Mohandessin, Giza', 7),
(110, 'Ear Nose Throat Clinic', '60 Ramses Street, Cairo', 8);

INSERT INTO Doctor (doctor_id, doctor_name, phone_number, address, department_id) VALUES
(201, 'Dr. Ahmed Hassan', '0101000201', '15 Nile Street, Dokki, Giza', 1),
(202, 'Dr. Mona Ibrahim', '0101000202', '40 Tahrir Street, Downtown Cairo', 1),
(203, 'Dr. Karim Adel', '0101000203', '11 Abbas El Akkad Street, Nasr City, Cairo', 2),
(204, 'Dr. Sara Nabil', '0101000204', '19 El Merghany Street, Heliopolis, Cairo', 2),
(205, 'Dr. Laila Samir', '0101000205', '30 Makram Ebeid Street, Nasr City, Cairo', 3),
(206, 'Dr. Omar Youssef', '0101000206', '33 Syria Street, Mohandessin, Giza', 4),
(207, 'Dr. Hany Fawzy', '0101000207', '63 Gameat El Dewal Street, Mohandessin, Giza', 5),
(208, 'Dr. Dina Farouk', '0101000208', '75 El Merghany Street, Heliopolis, Cairo', 6),
(209, 'Dr. Tamer Said', '0101000209', '16 El Batal Ahmed Street, Mohandessin, Giza', 7),
(210, 'Dr. Nadia Lotfy', '0101000210', '61 Ramses Street, Cairo', 8);

INSERT INTO Patient (patient_id, patient_name, phone_number, address, birth_date, job) VALUES
(12527, 'Mostafa Mahmoud', '01120012527', '8 El Gezira Street, Zamalek, Cairo', '1985-04-12', 'accountant'),
(12528, 'Farah Ghaly', '01120012528', '20 El Hegaz Street, Heliopolis, Cairo', '2001-09-21', 'Student'),
(12529, 'Youssef Ali', '01120012529', '13 Mostafa El Nahas Street, Nasr City, Cairo', '1994-01-15', 'Engineer'),
(12530, 'Mariam Samy', '01120012530', '44 Lebanon Street, Mohandessin, Giza', '1990-07-30', 'Teacher'),
(12531, 'Hassan Omar', '01120012531', '70 Faisal Street, Giza', '1978-11-03', 'Driver'),
(12532, 'Nour Khaled', '01120012532', '5 Makram Ebeid Street, Nasr City, Cairo', '1998-02-18', 'Pharmacist'),
(12533, 'Salma Ahmed', '01120012533', '3 Syria Street, Mohandessin, Giza', '1989-12-09', 'Designer'),
(12534, 'Omar Mostafa', '01120012534', '17 Ramses Street, Cairo', '2016-05-25', 'Student'),
(12535, 'Dalia Fathy', '01120012535', '14 Tahrir Street, Downtown Cairo', '1972-03-14', 'Banker'),
(12536, 'Khaled Nasser', '01120012536', '88 El Merghany Street, Heliopolis, Cairo', '1968-08-07', 'Retired');

INSERT INTO Appointment
(appointment_id, appointment_date, patient_id, doctor_id, start_time, end_time, cost, status, diagnosis) VALUES
(301, '2026-03-10', 12527, 203, '09:00:00', '09:30:00', 650.00, 'completed', 'Fatty liver'),
(302, '2025-09-18', 12528, 204, '10:00:00', '10:30:00', 600.00, 'completed', 'Fatty liver grade 1'),
(303, '2026-06-05', 12529, 201, '11:00:00', '11:30:00', 700.00, 'scheduled', 'Chest pain evaluation'),
(304, '2026-05-20', 12530, 202, '12:00:00', '12:30:00', 750.00, 'postponed', 'Hypertension follow-up'),
(305, '2024-11-15', 12531, 205, '13:00:00', '13:30:00', 400.00, 'completed', 'Eczema'),
(306, '2026-01-22', 12532, 206, '09:30:00', '10:00:00', 350.00, 'completed', 'Seasonal flu'),
(307, '2026-07-01', 12533, 207, '14:00:00', '14:45:00', 800.00, 'scheduled', 'Knee pain'),
(308, '2026-02-12', 12534, 208, '15:00:00', '15:30:00', 900.00, 'completed', 'Migraine'),
(309, '2025-12-03', 12535, 209, '16:00:00', '16:30:00', 500.00, 'completed', 'Myopia'),
(310, '2026-05-28', 12536, 210, '17:00:00', '17:30:00', 450.00, 'in progress', 'Sinusitis'),
```

TABLE	NUMBER OF RECORD
DEPARTMENT	10
DOCTOR	10
Patient	10
CLINIC	10
APPOINTMENT	15

THE TEST DATA INCLUDES:

- Patient ID 12527
- Cardiology department
- Cardiology clinics
- Fatty liver diagnosis records
- Different appointment statuses

TRIGGER

Trigger: Preventing Overlapping Appointments

A trigger was added to prevent the same doctor from having overlapping appointments on the same date.

The trigger checks:

* Doctor ID
* Appointment date
* Start time
* End time

If an overlapping appointment already exists, the new appointment is rejected.

Purpose

This prevents scheduling conflicts and improves data integrity.

FLASK APPLICATION

The application connects to the MySQL database and allows users to:

. View departments
* View clinics
* View doctors
* View patients
* View appointments
* Add new patients
* Add new appointments
* Search appointments
* View payment summaries

```
from flask import Flask, render_template, request, redirect, url_for, flash
import mysql.connector
import os

app = Flask(__name__)
app.secret_key = "clinic_system_secure_token"

def get_db_connection():
    return mysql.connector.connect(
        host=os.getenv("DB_HOST", "localhost"),
        port=int(os.getenv("DB_PORT", "3306")),
        user=os.getenv("DB_USER", "root"),
        password=os.getenv("DB_PASSWORD", ""),
        database=os.getenv("DB_NAME", "clinic_management_system")
    )

@app.route("/")
def index():
    db = get_db_connection()
    cursor = db.cursor(dictionary=True)
    cursor.execute("SELECT COUNT(*) AS total FROM Patient")
    patients_count = cursor.fetchone()['total']
    cursor.close()
    db.close()
    return f"Dashboard Live. Registered Patients Count: {patients_count}"

if __name__ == "__main__":
    app.run(debug=True, host="127.0.0.1", port=5000)
```

HOW TO RUN THE FLASK APP

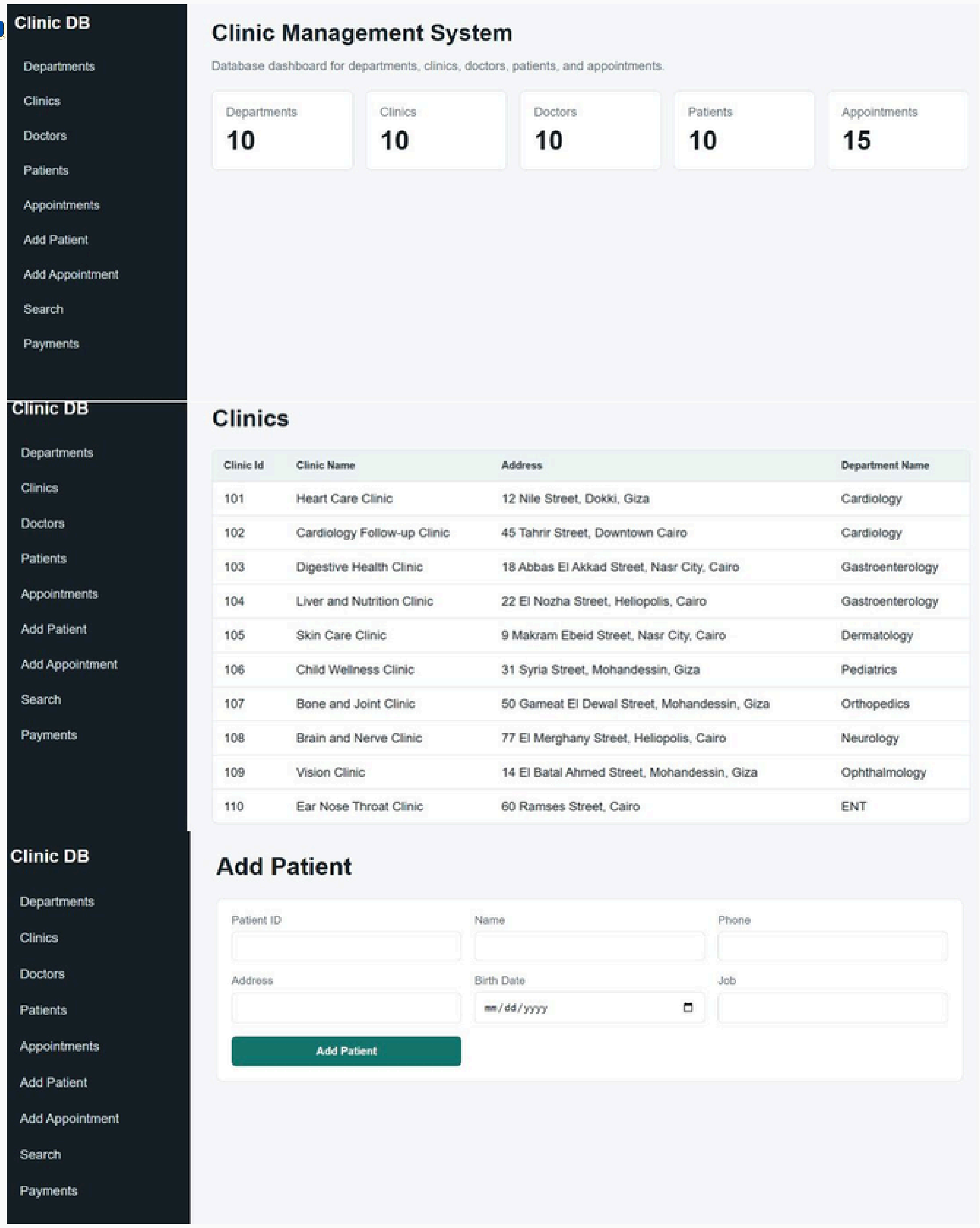
Steps:
1. Open the src folder
2. Create and activate the virtual environment
3. Install requirements
4. Run app.py
5. Open the local website

Commands:

```
bash
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
python app.py
```

Open in browser:
http://127.0.0.1:5000

1



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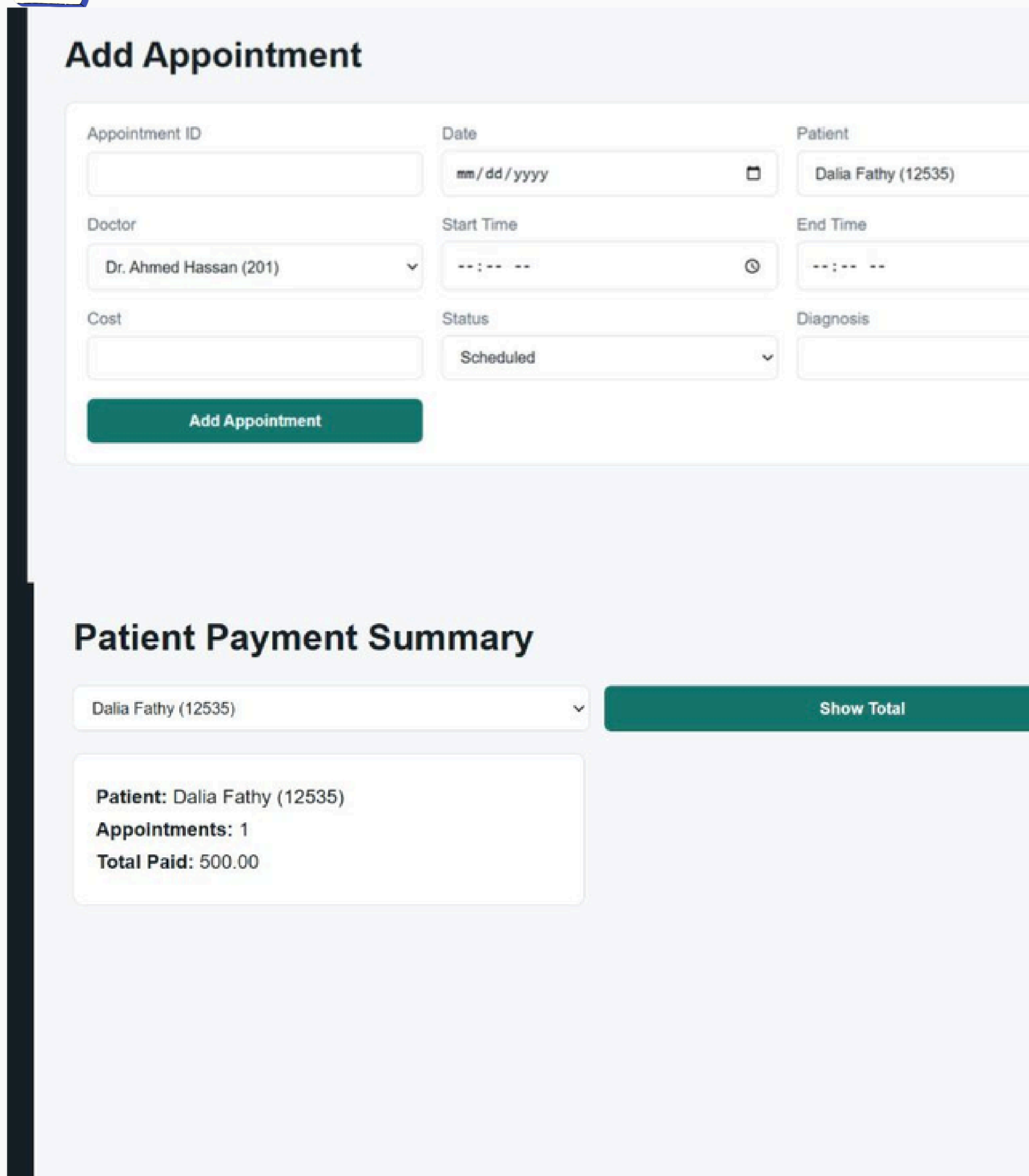
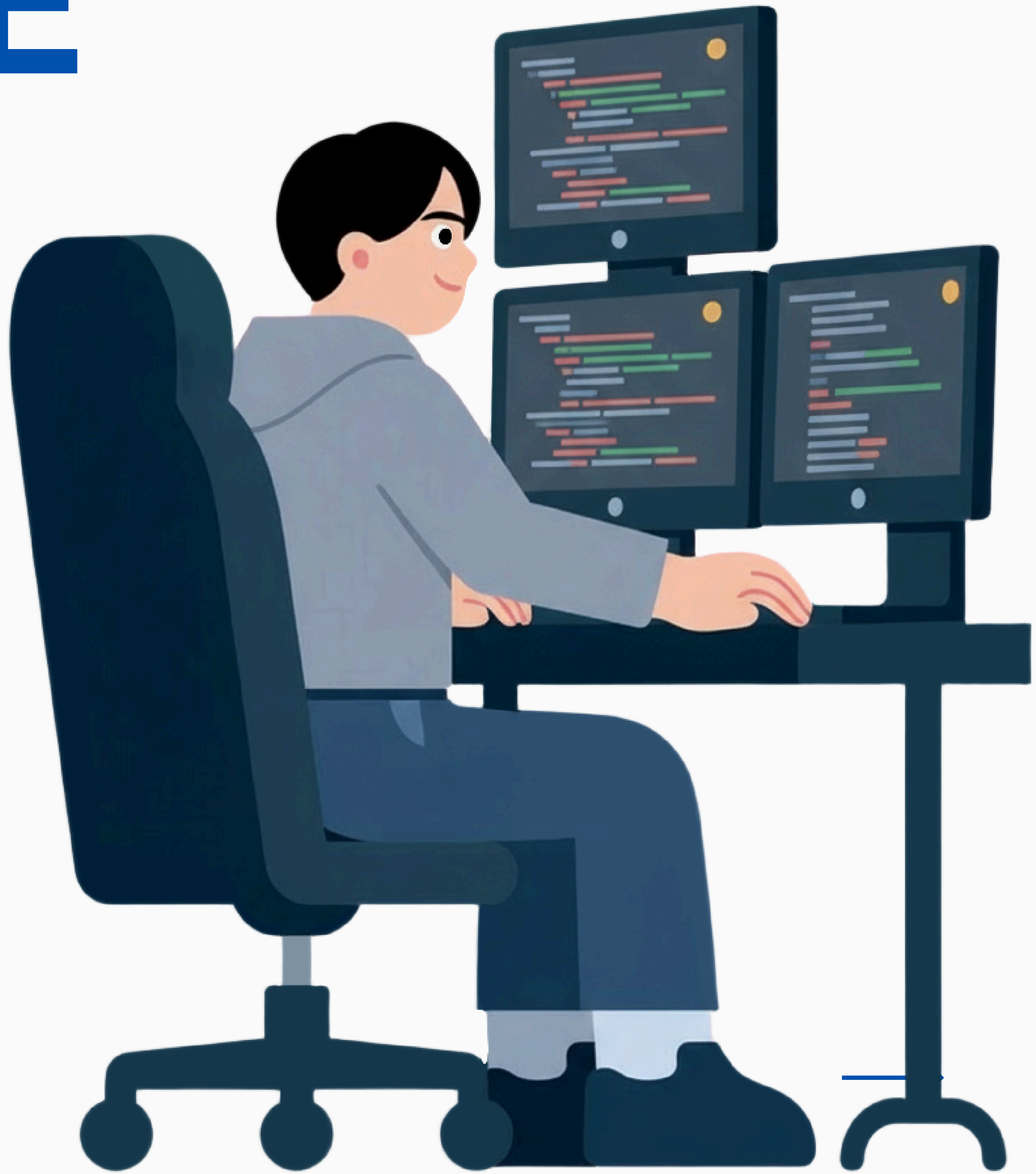


READ AND WRITE OPERATIONS

Read operations:
* Display departments
* Display clinics
* Display doctors
* Display patients
* Display appointments
* Display payment summaries

Write operations
* Add a new patient
* Add a new appointment



CONCLUSION

The Clinic Management System successfully implements a relational database for managing clinic operations.

The system supports:
* Department and clinic management
* Doctor and patient records
* Appointment scheduling
* Diagnosis tracking
* Payment summaries
* Doctor schedules
* Department revenue reports

THANK YOU SO MUCH!